**Week 8 In-Class Worksheet**
**Contingency Tables and Chi-Square Tests**

**Name:** ______________________ **Date:** ____________

**Course:** H524 Introduction to Biostatistics
**Topic:** Chi-Square Tests, Fisher's Exact Test, Effect Sizes

## Instructions

Work with a partner to solve the following problems. Show all your work including formulas, calculations, and interpretations. You may use R to verify answers, but demonstrate understanding by showing the steps.

## 1 Problem 1: Fisher's Exact Test

A small clinical trial tests a new treatment for a rare disease. Results:

| | Improved | Not Improved | Total |
|---|---|---|---|
| **Treatment** | 7 | 3 | 10 |
| **Control** | 2 | 8 | 10 |
| **Total** | 9 | 11 | 20 |

**Part A:** Calculate the expected counts for a chi-square test.

| | Improved | Not Improved |
|---|---|---|
| **Treatment** | | |
| **Control** | | |

**Part B:** Are the chi-square test assumptions satisfied? Explain.

Rule: All expected counts should be $\geq 5$

**Part C:** Based on your answer to Part B, which test should be used?

**Part D:** Use R to perform Fisher's exact test and report the p-value and odds ratio.

**R code:**

P-value (two-sided) =

Odds Ratio =

95% CI for OR =

**Part E:** State your conclusion about treatment effectiveness.

## 2 Problem 2: Chi-Square Test of Independence

A cohort study follows 500 smokers and 500 non-smokers for 10 years to assess heart disease incidence.

| | **Heart Disease** | **No Heart Disease** | **Total** |
|---|---|---|---|
| **Smoker** | 75 | 425 | 500 |
| **Non-smoker** | 35 | 465 | 500 |
| **Total** | 110 | 890 | 1000 |

**Part A:** State the hypotheses for a chi-square test of independence.

**Part B:** Calculate the expected counts for each cell.

Formula: $E_{ij} = \frac{(\text{Row } i \text{ total}) \times (\text{Column } j \text{ total})}{\text{Grand total}}$

| | **Heart Disease** | **No Heart Disease** |
|---|---|---|
| **Smoker** | | |
| **Non-smoker** | | |

**Part C:** Calculate the chi-square test statistic.

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} =$$

**Part D:** Find the p-value and state your conclusion. ($df = (r-1)(c-1)$)

$df =$

P-value =

Conclusion:

# 3 Problem 3: Effect Sizes - Relative Risk and Odds Ratio (OPTIONAL - Week 9 Preview)

**Note:** This problem previews next week's content on effect sizes. Try it for extra practice, but it will not be on this week's assignment.

A cohort study follows 500 individuals exposed to a chemical and 500 unexposed individuals for 5 years to assess disease incidence.

| | **Disease** | **No Disease** | **Total** |
|---|---|---|---|
| **Exposed** | 80 | 420 | 500 |
| **Unexposed** | 40 | 460 | 500 |
| **Total** | 120 | 880 | 1000 |

**Part A:** Calculate the relative risk (RR).

**Formula:** $RR = \frac{\text{Risk in exposed}}{\text{Risk in unexposed}} = \frac{a/(a+b)}{c/(c+d)}$

Risk in exposed group =

Risk in unexposed group =

Relative Risk (RR) =

**Part B:** Interpret the relative risk. What does this value tell us?

**Part C:** Calculate the odds ratio (OR).

**Formula:** $OR = \frac{a \times d}{b \times c}$ (cross-product ratio)

OR =

**Part D:** Compare RR and OR. Are they similar or different? Why?

**Part E:** Using R, verify your calculations and obtain a 95% confidence interval for the odds ratio.

**R code:**

95% CI for OR =

**Part F:** Based on the confidence interval, is there significant evidence of an association between exposure and disease? Explain.

## R Code Reference

**Fisher's exact test:**

```
data <- matrix(c(7, 3, 2, 8), nrow=2, byrow=TRUE)
fisher.test(data)
```

**Chi-square test of independence:**

```
data <- matrix(c(75, 425, 35, 465), nrow=2, byrow=TRUE)
chisq.test(data)
```

**Effect sizes (Week 9 Preview):**

```
# Relative Risk
risk_exposed <- 80/500
risk_unexposed <- 40/500
RR <- risk_exposed / risk_unexposed

# Odds Ratio with CI
data <- matrix(c(80, 420, 40, 460), nrow=2, byrow=TRUE)
fisher.test(data)
```